**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 118 · SQL Queries

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why open a file before reading it?

Q2. What does CPU cache hold?
A. Only saved personal files
B. Often-used data and instructions
C. Only website passwords
D. Only the next address

Q3. “A register is a large external drive used for backups.” Correct this. Explain why.

Q4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

SELECT — Identifies which field or fields a query retrieves.

FROM — Names the table containing the records.

WHERE — Filters rows using a condition.

SQL query — A database query retrieves only the requested information from a table.

### New learning

Field and record: A SQL table stores each pupil as a row/record; Name and Mark are columns/fields.

SELECT: Names the field(s) to return in the result, such as Name or ID.

FROM: Names the table to search; our shared example is Pupils.

WHERE: Filters rows. Comparing the numeric Mark to 50 returns only pupils meeting the condition.

### Worked example

Using Pupils(ID, Name, Mark): 101 Ada 72; 102 Ben 43; 103 Mia 81. Write SQL to return the names of pupils with Mark >= 50, and state the result.

1. SELECT Name: ask for the Name field, not the complete record.

2. FROM Pupils: identify the table holding these rows.

3. WHERE Mark >= 50: Ada (72) and Mia (81) match; Ben (43) does not.

4. SELECT Name FROM Pupils WHERE Mark >= 50; returns Ada and Mia, in no guaranteed order unless an ordering rule is specified.

## We Do · Practise with guidance

### Guided 1

Using the same Pupils table, write SQL to return the ID of pupils with Mark < 50. Identify the result. [4 marks]

Worked method: SELECT ID FROM Pupils WHERE Mark < 50; returns 102. SELECT/FROM/WHERE [3] and correct resulting row [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Write an SQL query returning Name where Class = 11.

```
SELECT Name FROM Pupils ...
```

write

### Question 1

Write SQL to return Name from Pupils where Mark > 70. [3 marks]

### Question 2

For the shared table, what value is returned by SELECT Name FROM Pupils WHERE ID = 103;? [2 marks]

### Question 3

Explain how SELECT and WHERE perform different jobs in SELECT Name FROM Pupils WHERE Mark < 50; [3 marks]

## Exit Ticket · Answer before checking

SQL WHERE selects the table name.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Opening establishes a handle or connection through which the program accesses the file.

Q2. B — Often-used data and instructions

Q3. A small, fast storage location inside the CPU. It holds data or addresses in use.

Q4. The accumulator holds the result, 18. The PC holds the next instruction's address, 240.

### I Do / We Do pages · Guided practice

Q1. SELECT ID FROM Pupils WHERE Mark < 50; returns 102. SELECT/FROM/WHERE [3] and correct resulting row [1].

### You Do pages · Code First

```
SELECT Name FROM Pupils WHERE Class = 11;
```

Expected result: names from matching rows (table supplied)

### You Do pages · Independent questions

Q1. SELECT Name [1] FROM Pupils [1] WHERE Mark > 70 [1]; result Ada and Mia.

Q2. Mia [1]; because the row with ID 103 matches the WHERE condition [1].

Q3. SELECT Name chooses the field to output [1]; WHERE Mark < 50 filters qualifying rows [1]; in this dataset only Ben matches [1].

### Exit Ticket · Model answer

False

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**